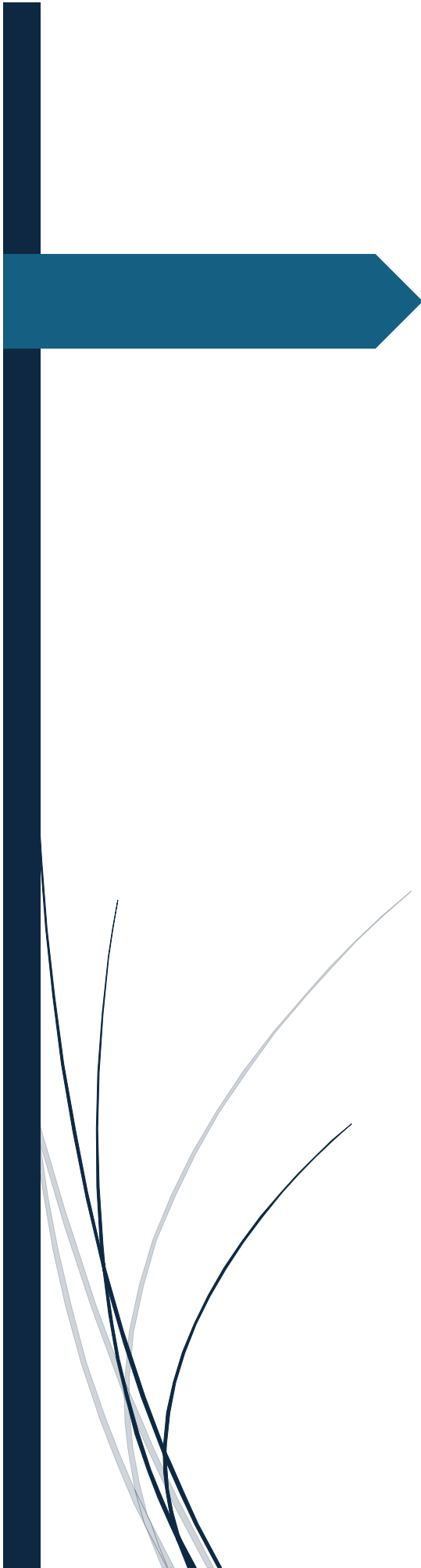




Assignment 2

Diagnosing Hardware Failures
Theory

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. RAM problem	PC does not start or keeps restarting
2. Hard disk problem	PC becomes very slow or files do not open
3. Power supply problem	PC does not turn on
4. Overheating	PC becomes hot and shuts down
5. Monitor problem	Screen is blank or shows no display

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

1. Check the charger

Make sure the charger is connected properly and the power light is on.

2. Check the power socket

Try another socket to make sure the socket is working.

3. Check the charging cable

Look for cuts, loose parts, or damage in the cable.

4. Check the charging port

Check if the laptop charging port is dirty, loose, or damaged.

5. Check the battery

If possible, remove the battery and check for damage or swelling.

6. Test with another charger

Use a compatible charger to see if the laptop starts charging.

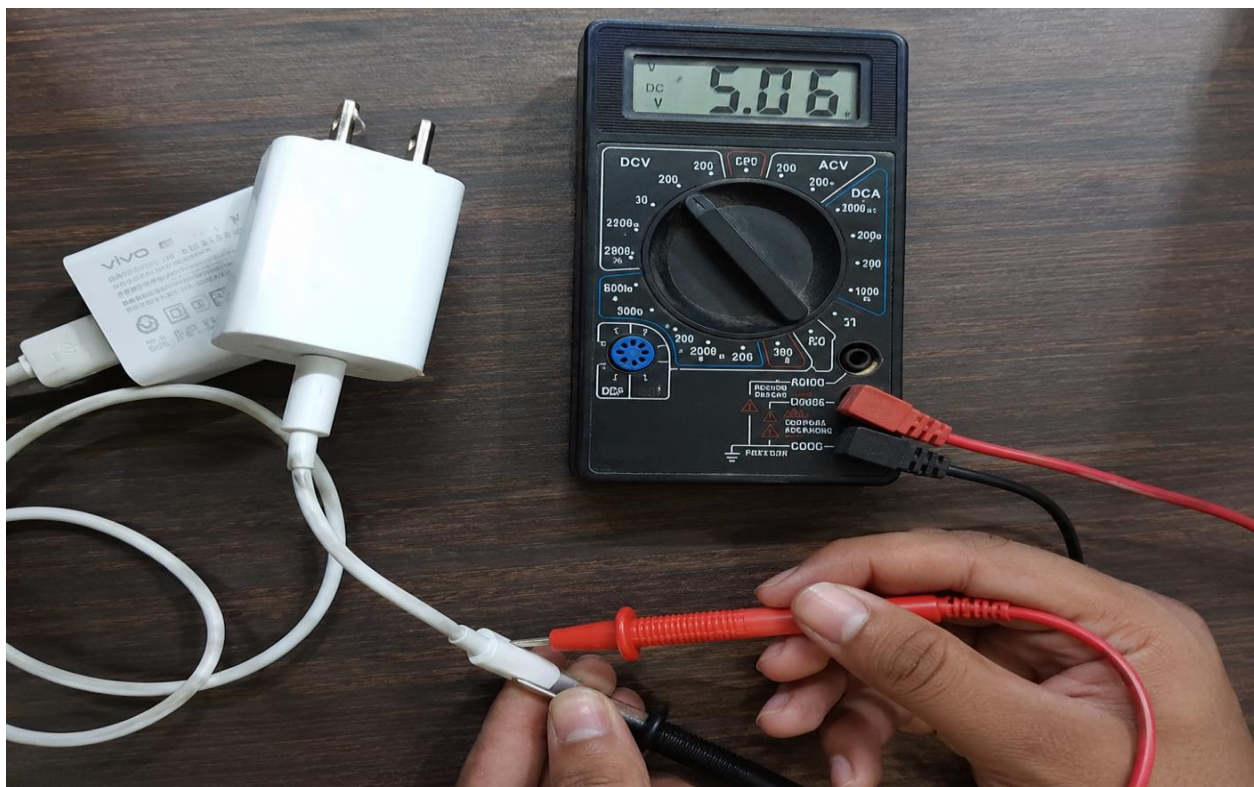
7. Check with a multimeter

Measure the charger's output voltage to see if it is giving power.

8. Final check

If the laptop still does not charge, the **battery, charger, or charging port** may need repair or replacement.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).



➤ The multimeter showed a reading of **5.06V**. The expected USB voltage range is **4.75V to 5.25V**, so the measured voltage is within the normal range. This means the charger is giving a **safe and normal voltage output**.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

1. Turn off the PC and unplug the power cable.
2. Disconnect the **ATX power connector** from the motherboard.
3. Set the multimeter to **DC voltage (V=)**.
4. Turn on the PSU safely without connecting it to the motherboard.
5. Put the **black probe on a black wire (GND)**.
6. Put the **red probe on the yellow wire (+12V)**.
7. Check the voltage shown on the multimeter.
8. The normal reading should be around **12V**, usually between **11.5V and 12.6V**.

Measured voltage: 12.08V

Result: The voltage is normal.

If the voltage is **below 11.5V**, it could indicate a **weak or faulty power supply (PSU)**. This may cause the PC to randomly shut down or restart.